

Platelet-rich plasma augmentation in anterior cruciate ligament reconstruction: Evidence is still too scattered. A scoping review of randomised controlled trials

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Abstract

Purpose: Platelet-rich plasma (PRP) augmentation has been proposed to improve the results of anterior cruciate ligament reconstruction (ACLR). The present study aims to quantify the available evidence to support the use of PRP as biological augmentation in ACLR surgery.

Methods: A systematic literature search was conducted on the PubMed, Cochrane, Web of Science and Embase databases on 10 March 2023. Inclusion criteria were randomised controlled trials (RCTs), written in English, addressing PRP augmentation in ACLR surgery, with no time limitation. A scoping review was performed to map the body of literature by examining the evidence related to specific aspects of patients' treatment and evaluation. Risk of bias evaluation was performed with the Cochrane risk-of-bias tool for randomised trials Version 2 (RoB 2), while the quality assessment was performed with the use of the Coleman Score.

Results: Out of 983 articles retrieved, 23 RCTs on 943 knees were included in this scoping review. PRP was administered in a liquid form in nine studies and clotted in 11 studies, while in three studies both liquid and clotted PRP were used. Hamstring auto/allografts were used in 14 studies, patellar tendon auto/allografts were used in eight studies and one study described ACLR with peroneus longus allografts. The map of the evidence documented high heterogeneity also in terms of surgical technique, objective and subjective outcome measures and radiological assessment, as well as follow-up times ranging from 1 day to 2 years, with virtually no overlapping data among studies neither in terms of treatments nor evaluations. Risk of bias evaluation showed an overall low quality of the included studies.

Conclusions: The available literature addressing PRP augmentation in ACLR is largely scattered. PRP was produced and applied following different procedures, and high variability was detected across the included studies for every aspect of ACLR surgery and evaluation. Currently, a

Abbreviations: ACL, anterior cruciate ligament; ACLR, anterior cruciate ligament reconstruction; ADC, apparent diffusion coefficient; DCE-MRI, dynamic contrast-enhanced magnetic resonance imaging; IKDC, International Knee Documentation Committee Knee Form; MRI, magnetic resonance imaging; PRF, platelet-rich fibrin; PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analysis; PRP, platelet-rich plasma; RCTs, randomised controlled trials; RoB 2, Cochrane risk-of-bias tool for randomised trials Version 2; VAS, Visual Analogic Scale; VISA, Victorian Institute of Sport Assessment.

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meaningful comparison of the available studies is not possible as the quantification of the literature results is biased by their heterogeneity. Future studies should provide more standardisation to investigate the benefits of biological augmentation in ACL surgery and confirm the promising yet weak evidence of PRP potential as well as the most suitable application modality, before routine use in clinical practice.

Level of Evidence: Levels I and II, scoping review.

KEYWORDS

anterior cruciate ligament, biological augmentation, platelet-rich plasma (PRP), reconstruction, scoping review

INTRODUCTION

Anterior cruciate ligament (ACL) tears are among the most common sport-related injuries and ACL reconstruction (ACLR) surgery is one the most frequently performed procedures in the field of orthopaedics and sport medicine [8]. The majority of patients presenting with ACL injury are young and sport-active and have high expectations in terms of functional recovery and return to sport [21]. Despite the overall good clinical results offered by the numerous different techniques developed over the years, ACLR is not a “100%-success” procedure [28, 57]. Moreover, ACL-injured patients are often a very demanding category, especially sport professionals who need to return to the playing field as soon as possible [17]. Nonetheless, only 55% of patients after ACLR return to the competitive level of sport [3].

Current research is investigating novel approaches to enhance ACL healing and accelerate recovery time with a higher activity level and a lower failure rate. To this aim, orthobiologics have been proposed as an augmentation strategy, with most of the research efforts being dedicated to investigating the potential of platelet-rich plasma (PRP) [37]. PRP exploits the high dose of growth factors, cytokines and bioactive molecules stored in platelet α -granules, which have been shown to take part in the regulation of joint homeostasis and the promotion of tissue healing, with immunomodulatory properties and anabolic effects [16, 56]. The healing potential of PRP has been supported by promising data from in vitro and preclinical in vivo studies, both in terms of safety and efficacy [7, 20]. PRP seems to promote a better and faster ligamentisation of the graft, to reduce the proinflammatory factors released immediately after surgery and to contribute to a better integration of the graft within the bone tunnels, thus avoiding their enlargement and failure over time [42]. Furthermore, PRP could be used to accelerate healing and reduce donor-site morbidity at the harvest site of the tendon graft, as well as to promote a healthier joint environment [1]. Accordingly, PRP augmentation is increasing in clinical practice, and a growing number of

studies have explored its use to optimise results in patients with ACLR.

The aim of the present study is to quantify all the available evidence to support the use of PRP as biological augmentation in ACLR surgery and to provide a comprehensive overview of different approaches, techniques, and clinical and functional outcomes of this new yet highly popular procedure. The hypothesis is that the available literature addressing PRP augmentation in ACLR is scattered and heterogeneous.

MATERIALS AND METHODS

A literature search was conducted on the PubMed, Cochrane, Web of Science and Embase databases on 10 March 2023 using the following criteria: (PRP OR platelet-rich plasma OR PRF OR platelet-rich fibrin OR plasma rich in growth factors OR platelet-derived growth factor OR platelet-derived OR platelet gel OR platelet concentrate OR autologous conditioned plasma OR autologous protein solution OR platelet lysate OR platelet supernatant) AND (ACL OR anterior cruciate ligament OR ACL reconstruction OR ACLR). The guidelines for Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) were used (Figure 1). Two independent observers (A. Be. and A. S.) conducted the screening process and analysis separately. First, the articles were screened by title and abstract. The following inclusion criteria were used: randomised controlled trials (RCTs), written in the English language, with no time limitation, on the use of PRP in ACLR. All perioperative PRP applications (graft soaked in PRP, PRP gel, intra-articular injections) and harvest site administration were included. Exclusion criteria were articles written in other languages, nonrandomised studies, preclinical studies, study protocols, congress abstracts, commentaries, reviews, meta-analyses or studies analysing other applications of PRP in knee surgery not related to ACLR procedures. In the second step, the full texts of the selected articles were screened, with further exclusions according to the previously described criteria. Moreover,

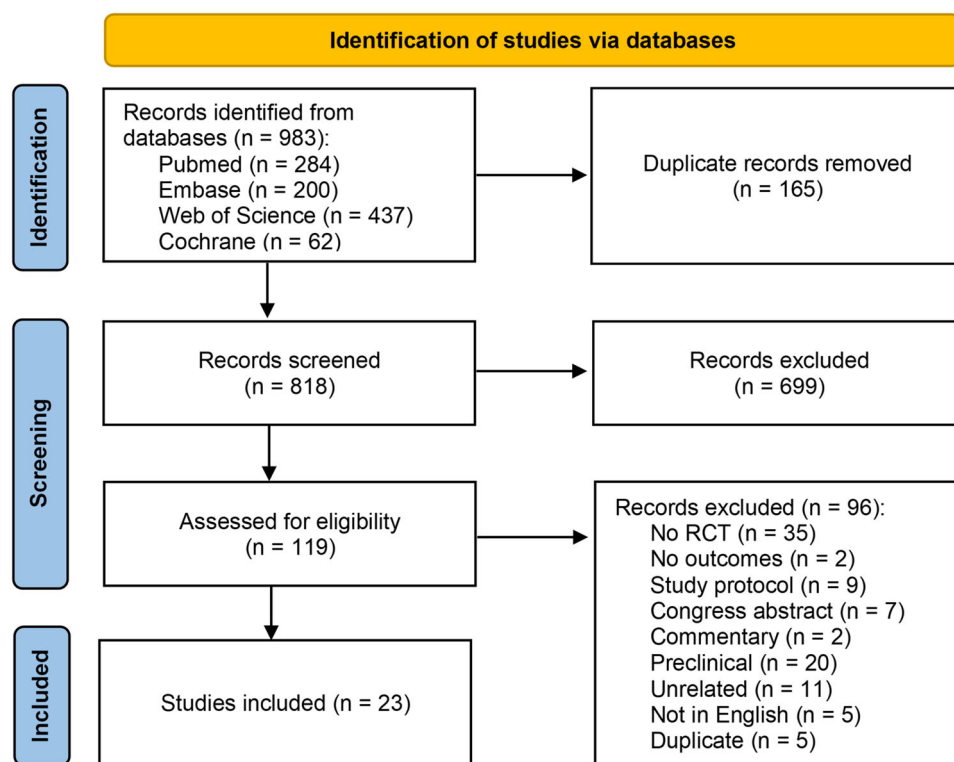


FIGURE 1 PRISMA flowchart. PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses; RCT, randomised controlled trial.

the articles not reporting clinical or radiological results were excluded. Relevant data (author, year of publication, number of patients, sex, age, follow-up time, PRP preparation and application modality, type of ACL graft used and clinical and radiological outcomes) were then extracted. Based on the systematic search of the literature to quantify the evidence supporting PRP augmentation to ACLR, a scoping review was also performed to determine the scope or coverage of the body of literature and give clear indications of the volume of literature and studies available as well as a detailed overview of its focus, thus providing a map of the available evidence.

Risk of bias and quality of evidence

The quality of each article was assessed independently by two authors (A. B. and A. S.) using the Cochrane risk-of-bias tool for randomised trials Version 2 (RoB 2) [55], while the quality assessment was performed with the use of the Coleman Score [12]. RoB 2 is structured into a fixed set of domains of bias, focusing on different aspects of trial design, conduct and reporting. Within each domain, a series of questions ('signalling questions') aim to elicit information about features of the trial that are relevant to the risk of bias. A proposed judgement about the risk of bias arising from each domain is generated by an algorithm, based on

answers to the signalling questions. Judgement can be 'Low' or 'High' risk of bias, or can express 'Some concerns'.

RESULTS

A total of 983 articles were retrieved. After the removal of the duplicates and screening of the titles, abstracts and full texts, 23 RCTs were included. Among these papers, six [5, 46, 47, 49, 50, 63] were follow-ups of previous papers and therefore referred to the same original patient series (Figure 2). Thus, this review included 17 series for a total of 943 knees. Three studies did not report patients' gender, the others reported the results in 645 men (80.2%) and 172 women (19.8%), with a mean age of 28.4 ± 8.2 years. Overall, 524 knees (79.3% men, 20.7% women) underwent ACLR with PRP augmentation, while 419 knees (81.6% men, 18.4% women) underwent conventional ACLR and were analysed as a control group. PRP was administered in a liquid form in nine studies and clotted (PRP clot/gel/platelet-rich fibrin—PRF) in 11 studies, while in three studies both liquid and clotted PRPs were used. Hamstring (HS) grafts were used in 14 studies, patellar tendon grafts were used in eight studies and one study described ACLR with peroneus longus allograft. Baseline characteristics of the

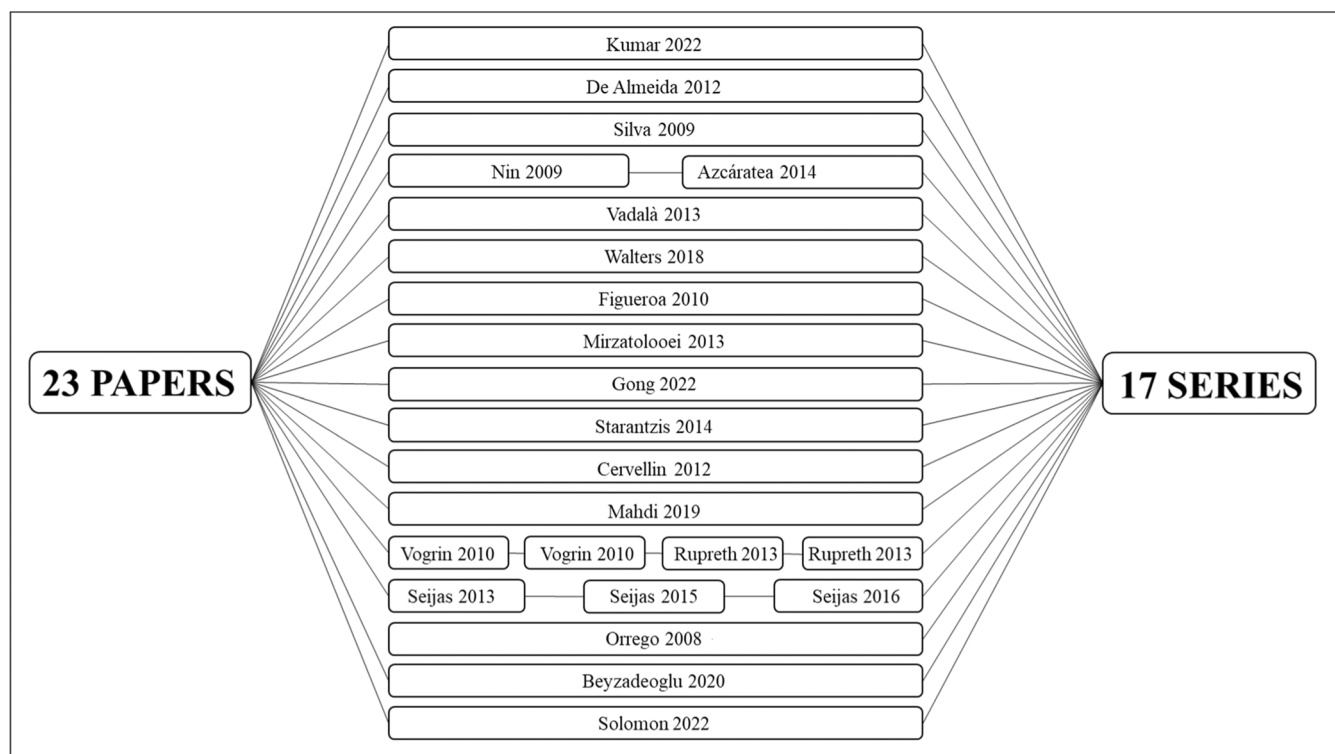


FIGURE 2 Correspondence between the 23 articles retrieved and the 17 series of patients analysed.

included patients and PRP preparation and application of the included RCTs are reported in Table 1.

Clinical outcomes

Clinical objective outcomes were assessed in 10 studies (nine series of patients) [5, 19, 22, 31, 35, 40–42, 53, 58, 62] with Pivot shift, Lachman, objective International Knee Documentation Committee Knee Form (IKDC) [26] and anterior drawer tests or arthrometers, such as KT-1000, KT-2000 and Roli-meter. Two studies documented a benefit of PRP augmentation: Mahdi et al. [35] reported significantly better results in the PRP group in terms of anterior drawer, Lachman and Pivot shift tests, while Vogrin et al. [62] documented significantly better KT-2000 values in the PRP group at 3 and 6 months follow-up. On the other hand, the other eight studies documented no advantages of PRP augmentation.

Clinical subjective outcomes were evaluated in 13 studies (12 series of patients) [5, 6, 10, 13, 22, 31, 35, 41, 42, 49, 53, 58, 64]. Questionnaires administered were the Lysholm Knee Scoring Scale [33], Tegner Activity Scale [34], subjective IKDC, Victorian Institute of Sport Assessment (VISA) [61] and pain Visual Analogic Scale (VAS). Four studies showed a benefit of PRP augmentation: Mahdi et al. [35] in terms of Lysholm score at three months of follow-up, Cervellin

et al. [10] in terms of VISA score at 12 months, and de Almeida et al. [13] and Seijas et al. [49] in terms of pain VAS in the immediate postoperative period and at 2 months follow-up, respectively. On the other hand, the other nine studies documented no advantages of PRP augmentation.

Adverse events were investigated by 12 studies, with one study reporting benefits from ACLR augmentation: the study of Beyzadeoglu et al. [6] on 44 patients documented that the application of PRF resulted in significantly lower rates of postoperative hemarthrosis that needed to be aspirated.

Imaging results

Twelve studies (11 series of patients) assessed graft maturation and integration into the bone tunnel via magnetic resonance imaging (MRI), with radiological follow-ups ranging from 4 weeks to 1 year post-operatively [5, 6, 19, 22, 35, 41, 42, 48, 51, 52, 63, 64]. In two of these studies [48, 63], multiple MRI follow-ups were performed, while for the remaining studies only one MRI was performed. Six studies documented significantly better results in the PRP group in terms of graft maturation, integration and signal intensity [6, 35, 42, 48, 52, 63], while the remaining six studies documented no advantages of PRP augmentation.

TABLE 1 Characteristics of the included RCTs.

References	Treatment	Number	Sex, M/F	Age	PRP preparation	PRP application	PRP type	Graft	Conclusions
[31]	PRP	35	22/13	28.34 ± 4.32	Freshly prepared PRP (rapid single spin method)	Injected in grafts and tunnels at the end of the surgery	L	HS	No difference in Lysholm score and tibial tunnel widening improvements among the two groups
	Control	35	23/12	29.71 ± 2.99					
[13]	PRP	12	10/2	25.8	Haemonetics MCS1 9000 cell separator (Haemonetics Corp.) Platelets: 1 185 166/mm ³ WBC: 0.91/mm ³	Patellar tendon defect was completely filled with 20–40 mL of PRP gel	C	BPTB	PRP reduced postoperative pain At 6 months, PRP had a positive effect on patellar tendon harvest site healing on MRI. No difference in questionnaire and isokinetic testing results
	Control	15	14/1	23.1					
[51]	PRP at surgery	10	38/2	26.8 ± 5.3	PRP: 3 mL obtained from 27 mL of autologous blood and 3 mL of citric acid. 15 min of centrifugation at 3200 rpm	1.5 mL of PRP between the strands of the graft in each femoral tunnel	C	HS	PRP, isolated or with thrombin, does not accelerate tendon integration
	PRP + intra-articular after surgery	10			Thrombin: 6 mL obtained from 12 mL of autologous citrated blood				
	PRP activated with thrombin	10							
	Control	10							
[5]	Single-spinning PRGF	50	43/7	27.4	40 mL of blood was collected 1 h prior to surgery into sodium citrate tubes 3.8%	Graft was covered with PRP gel and sutured over itself with gel in its interior + gel in the tibial tunnel	C	P	PRGF reduced swelling; however, graft intensity and uniformity on MRI were similar in the three groups, and there was no clinical or pain improvement compared with the control group
	Double-spinning PRP with leucocytes	50	40/10	26.1	Single spinning: Centrifugation at 580 g for 8 min at 1800 rpm. Platelets: 242 × 10 ⁶ /mL				
	Control	50	38/12	26.1	Double-spinning: 8 min at 3000 rpm and 6 min at 1000 rpm Platelets: 837 × 10 ⁶ /mL				
[58]	PRP	20	20/0	34.5	Centrifugation of 10 mL of blood collected 1 h prior to surgery	5 mL between graft and tunnel	L + C	HS	PRP is not effective in preventing or even reducing femoral and tibial tunnel enlargement
	Control	20	20/0		Gel obtained with the addition of thrombin and 10% Ca-gluconate a few minutes before its use	5 mL of semisolid PRP above the graft 5 mL of liquid and semisolid PRP before fixing the graft			
[64]	PRP	27	NR	30 ± 12	10 mL blood + PRP separation kit and centrifuge system (ACP PRP; Arthrex)	Activated with 0.25 mL of CaCl ₂ , mixed with autologous cancellous bone chips and placed into the patellar donor site	C	BPTB	PRP has no significant effect on postoperative knee pain, IKDC scores or donor site healing
	Control	23							

(Continues)

TABLE 1 (Continued)

References	Treatment	Number	Sex, M/F	Age	PRP preparation	PRP application	PRP type	Graft	Conclusions
[19]	APC	30	18/12	26.8	55 mL of blood with 5 mL of citrate. Centrifuged for 15 min at 3200 rpm	3 mL tibial tunnel	C	HS	APC has no significant effect in terms of integration and graft maturation (ligamentisation)
	Control	20	15/5	23.6		3 mL femoral tunnel 4 mL in the intra-articular portion of the graft			
[40]	PRP	23	20/3	26.4	10 mL of the patient's blood was centrifuged at 1500 rpm for 5 min, from which 3.5 mL of PRP was obtained	Graft was immersed in PRP solution + 3 mL of PRP injected into the femoral and tibial tunnels	L	HS	PRP has no significant effect in preventing tunnel widening
	Control	23	22/1	26.9					
[22]	PRP	30	18/12	33.5 ± 8.97	4 mL obtained from 36 mL of blood + 4 mL anticoagulant citrate dextrose	Gelatin sponge filled with PRP affixed to the graft	L	HS	PRP has no significant effect on bone tunnel widening, tendon bone healing or knee function
	Control	30	21/9	34.9 ± 9.68					PRP may improve graft maturation
[41]	PDGF	50	38/12	26.1	Double-spinning: 8 min at 3000 rpm and 6 minutes at 1000 rpm Platelets: $837 \times 10^6/\text{mL}$	Graft was covered with PRP gel and sutured over itself with gel in its interior + gel in the tibial tunnel	C	P	PDGF has no clinical or biomechanical effect
	Control	50	40/10	26.6					
[53]	PRP	25	38/13	29.4 ± 7.3	PRP: Centrifuged for 15 min at 3200 rpm 6 mL of PRP was collected Thrombin: 10 mL of patient's blood centrifuged at 200 rpm for 5 min and mixed with CaCl_2	3 mL between the strands of the graft + 3 mL injected into the femoral tunnel	L + C	HS	PRP can potentially eliminate the biologic triggers of tunnel enlargement
	Control	26		31.3 ± 8.0					
[10]	PRP	20	20/0	22.7 ± 3.5	PRP: 54 mL of blood + citrate centrifuged for 15 min at 3200 rpm Thrombin: 10 mL of venous blood + anticoagulant Centrifuged for 2 min at 3200 rpm + CaCl_2 solution PRP and thrombin mixed and left to gelified	PRP gel applied to the patellar and tendon bone plug harvest site	C	BPTB	PRP group showed a significant improvement in terms of pain reduction
	Control	20	20/0	22.9 ± 4.3					
[35]	PRP	13	13/0	25.77	100–150 mL of PRP 6 mL pouch for injection	At the end of the surgery, injection of inactivated PRP, 3 mL in the femoral tunnel and 3 mL intra-articularly	L	HS	PRP has beneficial effects on graft-bone integration at 12+ weeks follow-up
	Control	14	14/0						

TABLE 1 (Continued)

References	Treatment	Number	Sex, M/F	Age	PRP preparation	PRP application	PRP type	Graft	Conclusions
[47]	PRP	21	13/8	37.2 ± 8.4	Autologous PRP mixed with activated autologous thrombin	1 mL in each tunnel	C	HS	PRP group showed enhanced cortical bone formation encircling the tibial tunnel at 2.5 and 6 months follow-ups
	Control	20	15/5	32.6 ± 12.3		3 mL onto the graft			
[46]	PRPG	21	13/8	37.2 ± 8.4	Autologous PRP mixed with activated autologous thrombin	1 mL in each tunnel	C	HS	PRPG reduced oedema during the first postoperative month and increased vascular density and microvessel permeability in the proximal tibial tunnel at 1 and 2.5 months follow-up
	Control	20	15/5	32.6 ± 12.3		3 mL onto the graft			
[63]	PG	21	13/8	37.2 ± 8.4	52 mL of patient's blood + 8 mL 10% calcium citrate. Platelets: $190 \times 10^9/L$	Applied into femoral and tibial tunnels and the graft	C	HS	Locally applied PG enhanced early revascularisation of the graft in the osteoigamentous interface zone
	Control	20	15/5	32.6 ± 12.3					
[62]	PLG	22	13/9	35.4 ± 10.0	52 mL of patient's blood + 8 mL 10% calcium citrate Platelets: $192 \times 10^9/L$	Applied into femoral and tibial tunnels and the graft	C	HS	Locally applied PLG can improve knee stability
	Control	23	17/6	33.0 ± 12.5					
[50]	PRGF	23	NR	NR	Technique described by Anitua et al. [2]	Injected into the patellar and tibial bone gap	L	P	PRGF accelerated tissue regeneration processes
	Control	20				2 cc injected under the paratenon			
[49]	PRGF	23	20/3	NR	Technique described by Anitua et al. [2]	Injected into the patellar and tibial bone gap	L	BPTB	PRGF enabled faster remodelling of patellar tendon grafts
	Control	20	17/3			2 cc injected under the paratenon			
[48]	PRGF	49	NR	NR	8 mL obtained in the surgery room with PRGF technique (BTI Systems)	Injected into the suprapatellar joint after suture	L	P	PRGF decreased donor site pain
	Control	49							
[42]	PC	26	NR	30	67 mL autologous blood + anticoagulant agent using Biomet GPS II kit (Biomet).	5 mL of clotted PC added to the graft	L + C	HS	PC had an enhancing effect on graft maturation. No effect in osteoigamentous interface or tunnel widening evolution
	PC + bone plug	27				1 mL injected between the strands of the graft			
	Control	27			Centrifuged for 5 min at 200 rpm				
[6]	PRF	23	18/5	21.7	120 mL autologous blood centrifuged with a Vivostat device	Sprayed over the graft, inside the femoral tunnel and to areas where bleeding was expected	L	HS	PRF led to superior graft integration and maturation in the proximal third of the graft and significantly lower rates of postoperative hemarthrosis
	Control	21	17/4	22.1					

(Continues)

TABLE 1 (Continued)

References	Treatment	Number	Sex, M/F	Age	PRP preparation	PRP application	PRP type	Graft	Conclusions
[52]	PRP + demineralised bone matrix	13	8/5	32.7 ± 10.5	Blood drawn prepared utilising the Arthrex ACP® system. PRP was mixed with 5 mL of StimuBlast® (Arthrex)	1–2 mL injected into each tunnel + additional injection at the graft–bone interface	L	Peroneus longus	PRP + demineralised bone matrix provided a significantly lower prevalence of ganglion cyst, reduction of tunnel widening, improved graft maturation and decreased size of ganglion cyst
	Control	14	7/7	32.7 ± 12.7					

Abbreviations: APC, autologous platelet concentrate; BPTB, bone–patellar tendon–bone; C, clotted; CaCl₂, calcium chloride; HS, hamstring; IKDC, International Knee Documentation Committee Subjective Knee Form; L, liquid; M/F, male/female; MRI, magnetic resonance imaging; NR, not reported; number, number of patients; P, patellar; PC, platelet concentrate; PDGF, platelet-derived growth factor; PG, platelet gel; PLG, platelet-leucocyte gel; PRF, platelet-rich fibrin; PRGF, plasma rich in growth factors; PRP, platelet-rich plasma; PRP-G, platelet-rich plasma gel; rpm, revolutions per minute.

Eleven of the included RCTs (nine series of patients) analysed tibial and/or femoral bone tunnels with MRI [5, 22, 41, 42, 46, 47, 52, 53] (four studies with serial evaluations [42, 46, 47, 53]), computed tomography (CT) scan [22, 40, 58] and X-ray [31]. Two studies documented a significantly reduced tunnel widening in the PRP group [52, 53], and one series documented increased cortical bone formation, increased vascular density and reduced oedema in the tibial tunnel [46, 47]. On the other hand, the remaining seven studies documented no advantages of PRP augmentation.

Four studies using autologous patellar tendon grafts assessed donor site healing with MRI [10, 13, 64] or ultrasound [50]. Two studies found a significantly better improvement in the PRP group in terms of residual patellar gap area [13, 50], while the other two studies documented no advantages of PRP augmentation.

Risk of bias

The evaluation using the RoB 2 tool showed the low quality of the studies, with six papers falling in the ‘Low risk’ category, seven in the ‘Some concerns’ category and 10 in the ‘High risk’ category (Figure 3). Notably, six out of the 10 ‘High risk’ studies documented significant benefits of PRP augmentation. The quality assessment performed with the use of the Coleman Score documented an overall medium quality of the included studies. In fact, only two studies obtained a score ≥85, 13 studies had a score between 70 and 84 and eight studies reached a score between 50 and 69 (Figure 3).

Scoping review results

The purpose of conducting a scoping review was to identify and map the available evidence in relation to specific aspects, ranging from ACLR technique and PRP application, as well as outcome in terms of imaging findings, objective measurements, subjective results and adverse events (Figure 4). More in detail, a subanalysis of the available evidence has been performed with respect to each one of these factors (Table 2).

ACLR technique

With regard to the graft choice, HS tendons, patellar tendons and peroneus longus tendons were used in 14, eight and one study, respectively (Figure 5). Both autografts and allografts were employed. ACLR with HSs was performed both with single- or double-bundle

Authors	D1	D2	D3	D4	D5	Overall	Coleman Score
Gong 2022							55
Kumar 2022							60
Solomon 2022							63
Beyzadeoglu 2020							63
Mahdi 2019							66
Walters 2018							67
Seijas 2016							67
Seijas 2015							67
Azcáratea 2014							71
Starantzis 2014							73
Mirzatoioei 2013							74
Ruprecht 2013							74
Seijas 2013							75
Vadalá 2013							75
Vogrin 2013							76
Cervellin 2012							76
de Almeida 2012							78
Ruprecht 2012							79
Figuerola 2010							81
Vogrin 2010							82
Nin 2009							82
Silva 2009							85
Orrego 2008							88

FIGURE 3 Cochrane risk-of-bias tool for randomised trials Version 2 (RoB 2) and Coleman Score. RoB 2: green = low risk; yellow = some concerns; red = high risk.

techniques in four and one study, respectively, while it was not documented in nine studies. Different fixation methods were employed: in the studies specifying femoral fixation, cortical buttons were used in four studies, cross pins in 10 studies, tightropes in two studies, swing bridge in one study and interference screws in four studies. For tibial fixation, interference screws were used in 16 studies, pins in one study and tightropes in two studies.

PRP augmentation

PRP was used either in liquid (nine studies) or solid form (11 studies), with three studies employing both application modalities [42, 53, 58] (Figure 5). Out of the liquid applications, three studies applied PRP during ACLR [22, 40, 52] and six afterwards (five at different moments at the end of surgery, one study both at the end of surgery and after 2 weeks), with applications being either into the tunnels or intra-articularly [6, 31, 35, 48–50]. The application of PRP in solid form also varied greatly, both in terms of site and delivery modality: nine studies documented the application in the tunnels and around the graft [5, 19, 41, 46, 47, 51, 53, 62, 63], while in three studies clotted PRP was applied in the patellar tendon harvest site [10, 13, 64]. In two studies, PRP delivery was combined with a scaffold [22, 64]. PRP also differed in terms of preparation modality: the studies documented variable PRP formulations in terms of volume of blood

harvested, time and speed of centrifugation, activation and so forth (Table 1). PRP composition was documented only in five studies [5, 13, 41, 62, 63].

Imaging evaluation

Among the 12 studies (11 series of patients) which assessed graft maturation and integration [5, 6, 19, 22, 35, 41, 42, 48, 51, 52, 63, 64], different MRI techniques were employed, including plane MRI, contrast-enhanced MRI, dynamic contrast-enhanced magnetic resonance imaging (DCE-MRI) and diffusion-weighted imaging, with a follow-up ranging from 4 weeks to 1 year. Graft maturation and integration were analysed with the use of different MRI parameters, such as signal intensity, signal-to-noise ratio and apparent diffusion coefficient (ADC). Graft signal intensity was measured only in the femoral tunnel portion in three studies [35, 42, 51], and only in the intra-articular portion of the graft in one study [19]. It was used to evaluate graft maturation in four studies [6, 19, 42, 52], graft ligamentisation in two studies [19, 48], graft integration in the femoral and tibial tunnel in four studies [6, 19, 47, 51] and presence of synovial fluid at graft–tunnel interface in one study [6]. Signal-to-noise ratio was used to evaluate either graft maturation [6, 22, 52], graft integration [51] and graft revascularisation [63], and ADC was employed to evaluate oedema, vascular density and microvessels permeability of the graft [46].

Eleven RCTs (nine series of patients) analysed tibial and/or femoral bone tunnels. For this purpose, multiple imaging techniques were employed, including MRI [5, 22, 41, 42, 46, 47, 52, 53], CT scan [40, 58] and radiograph [31]. The timing of imaging acquisition at follow-up ranged from 6 weeks to 14.7 months postoperatively. One study analysed tibial tunnel widening alone [31], two studies analysed femoral tunnel widening alone [42, 53], while only three studies reported both tibial and femoral tunnel widening [40, 52, 58]. Two studies analysed cortical bone formation, oedema and vascular density in the tibial tunnel portion [46, 47].

Finally, patellar tendon donor site healing was assessed in three studies with MRI [10, 13, 64] and in one study with ultrasound [50].

Clinical objective outcomes

Among 10 studies that statistically compared anterior knee laxity between groups [5, 19, 31, 35, 40–42, 53, 58, 62], seven different tools were used: anterior drawer, Lachman test, Pivot shift test, objective IKDC, KT-1000, KT-2000 and Rolimeter, with a follow-up ranging from 6 weeks to 3 years. Of the two studies reporting Lachman test results, one [31] stratified Lachman test results in grade 1, 2 or 3 and the other

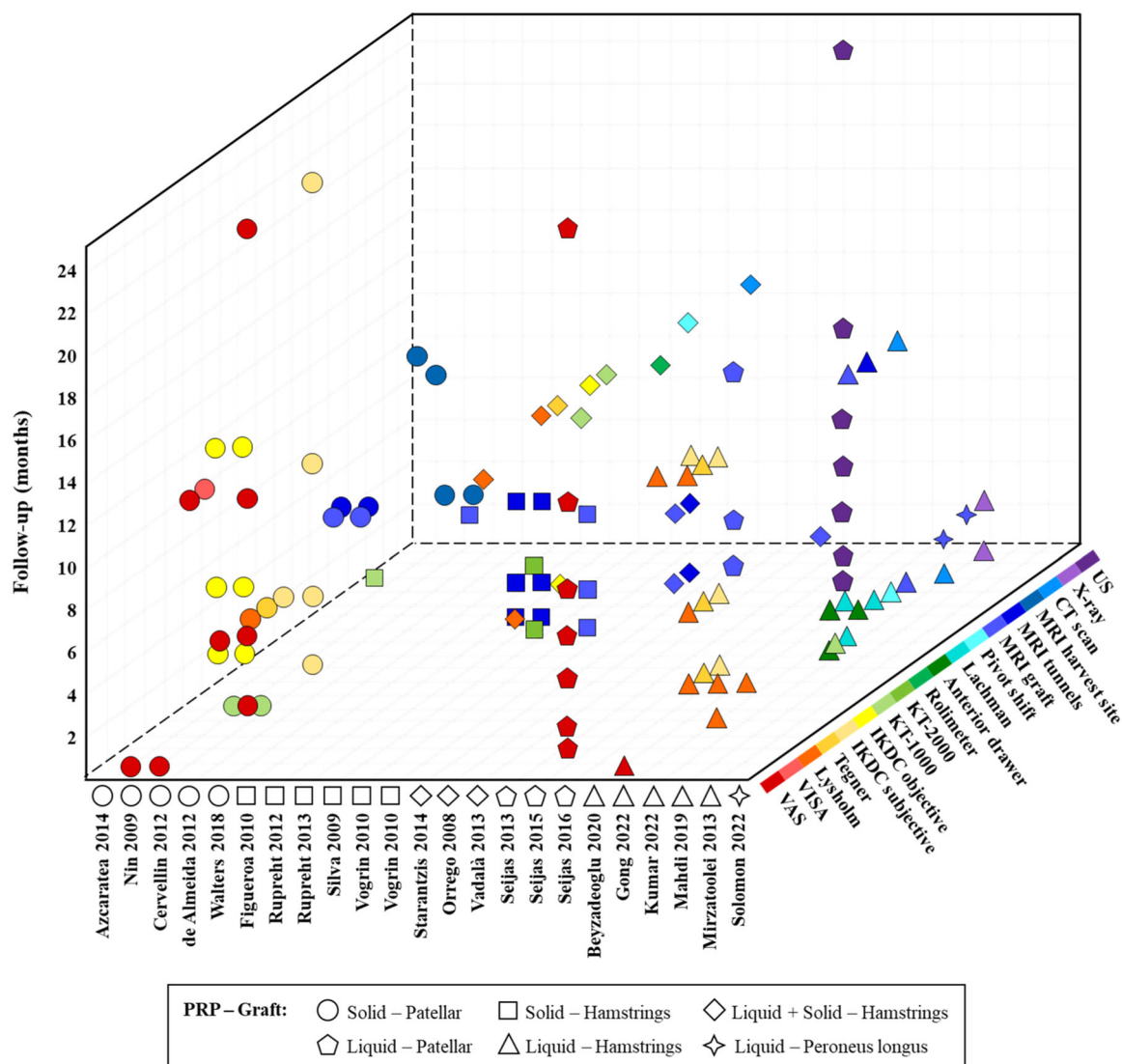


FIGURE 4 Three-dimensional representation of the outcomes at different follow-ups evaluated by the 23 studies included. PRP, platelet-rich plasma. CT, computed tomography; IKDC, International Knee Documentation Committee Knee Form; MRI, magnetic resonance imaging; US, ultrasound; VAS, Visual Analogic Scale; VISA, Victorian Institute of Sport Assessment.

one [35] described only two possible results (<5 mm or >5 mm), thus making a comparison impossible. Pivot shift test was performed in four studies. Of the two studies reporting anterior drawer test results, one stratified the results in accordance to the anterior tibial dislocation (<5, 5–10 and >10 mm) [35], while the other one described only grade 1, 2 or 3, without additional information [31]. Among the five studies reporting KT-1000 values, one study documented only postoperative values [19] and one study reported KT-2000 values [62]. Similarly, only one study reported Rolimeter values [53]. Four studies reported objective International Knee Documentation Committee Knee Form values (three series of patients): while three studies stratified the results in A, B, C and D categories, one study reported only two categories (A + B and C + D) [42].

Clinical subjective outcomes

Clinical subjective outcomes were evaluated in 13 studies through five different questionnaires: VAS pain, VISA score, Lysholm score, Tegner Activity Scale and subjective International Knee Documentation Committee Knee Form score with follow-ups ranging from the day after surgery to 2 years [5, 6, 10, 13, 22, 31, 35, 41, 42, 49, 53, 58, 64]. Five of the seven studies reporting VAS pain [5, 10, 13, 22, 41, 49, 64] were focused on patients who underwent ACLR with patellar tendon graft and the application of clotted PRP. Only one study reported the VISA score [10] to assess the disability of the patellar donor site after ACLR with bone–patellar tendon grafting technique. The Tegner Activity Scale was used in four studies. Eight studies reported the Lysholm score, with two studies reporting only

TABLE 2 Outcomes evaluated across the 23 included studies.

Group (PRP—Graft)	References	Clinical subjective					Clinical objective					Radiological							
												MRI							
		VAS	VISA	Lysholm	Tegner	Subjective IKDC	Objective IKDC	KT-1000	KT-2000	Rollimeter	Anterior drawer	Lachman	Pivot shift	Graft maturation/ integration	Tunnels	Patellar harvest site	CT scan	X-ray	US
Solid—Patellar	[5]	Yellow	Yellow				Yellow						Yellow	Yellow					
	[41]						Yellow												
	[10]		Green													Yellow	Green	Yellow	
	[13]	Green		Yellow	Yellow								Yellow	Yellow		Green			
	[64]	Yellow		Yellow	Yellow									Yellow		Yellow			
Solid—Hamstring	[19]							Yellow											
	[46]														Green				
	[47]														Green				
	[51]												Yellow	Yellow					
	[62]												Green	Green					
Liquid and solid—Hamstring	[63]								Green					Green					
	[53]			Yellow	Yellow		Yellow			Yellow					Green	Yellow			
	[42]			Yellow	Yellow		Yellow						Green	Green				Yellow	
	[58]			Yellow	Yellow	Yellow						Yellow		Green				Yellow	
	[48]			Yellow	Yellow									Green					
Liquid—Patellar	[50]																		Green
	[49]	Green																	
Liquid—Hamstring	[6]			Yellow	Yellow	Yellow								Green					
	[22]	Yellow		Yellow	Yellow	Yellow							Yellow	Yellow			Yellow		
	[31]			Yellow	Yellow	Yellow						Yellow						Yellow	
	[35]			Yellow	Yellow	Yellow						Green	Green	Green					
	[40]			Yellow	Yellow	Yellow								Green				Yellow	
Liquid—Peroneus longus	[52]												Green	Green	Green	Green	Green	Green	Green

Note: Green = better results in the PRP augmentation group; yellow = no difference between the PRP augmentation group and control group; white = outcome not reported.
Abbreviations: CT, computed tomography; IKDC, International Knee Documentation Committee Knee Form; MRI, magnetic resonance imaging; PRP, platelet-rich plasma; VAS, Visual Analogic Scale; VISA, Victorian Institute of Sport Assessment.

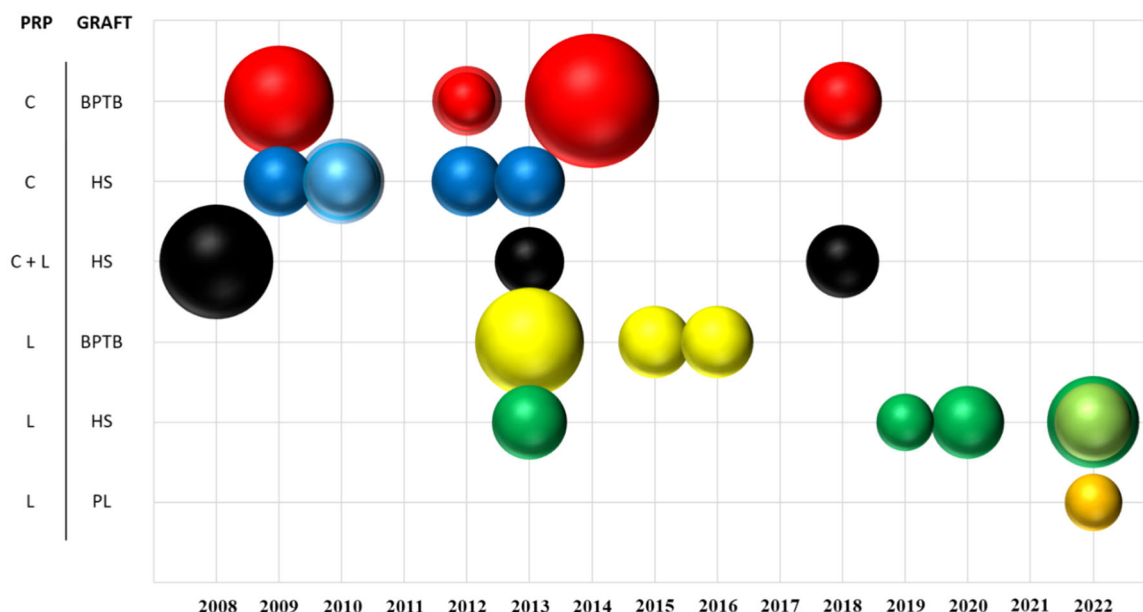


FIGURE 5 Platelet-rich plasma formulation and graft choice analysis. Sphere size represents the number of patients. BPTB, bone–patellar tendon–bone; C, clotted; HS, hamstring; L, liquid; PL, platelet–leucocyte; PRP, platelet-rich plasma.

postoperative Lysholm scores with no comparison to the presurgery patient status [6, 35]. The subjective IKDC score was reported by four studies, two of which did not report preoperative scores but only postoperative values [6, 64].

DISCUSSION

The main finding of this study is that the evidence on PRP augmentation in ACLR is still too scattered to draw clear indications on its use in clinical practice.

The interest of the scientific community around PRP has largely increased during the last years to accelerate and improve the results of treatments for tissues with low healing potential. This corresponded to a simultaneous growth in the clinical application and in scientific publications, including those on ligament pathologies such as ACLR. With the aim of summarising the growing body of evidence, many systematic reviews have been published on this topic, and recently even meta-analyses in the attempt to quantify the potential benefits of this biological augmentation [14, 32, 39, 65]. In 2021, the meta-analysis of de Andrade et al. [14] focusing on nine studies concluded that PRP showed no improvement in objective outcomes like ligamentisation and less tunnel widening, while it showed small improvements in terms of Lysholm, VAS and knee laxity. In the same year, by including a larger number of 17 RCTs, Lv et al. [32] found moderate quality of evidence suggesting that PRP could provide short-term pain reduction. More recently, the meta-analysis of McRobb et al. [39] on 13

RCTs concluded that PRP influences healing through early vascularisation, culminating in higher rates of ligamentisation, and the meta-analysis of Zhu et al. [65] on 14 studies confirmed that PRP applied alongside ACLR could reduce postoperative pain and improve knee function in the short and medium terms.

This review on a higher number of high-level studies, 23 RCTs, questions the findings of those publications. The systematic review of Kon et al. encompassed a higher number of articles but included an even larger number of heterogeneous products ranging from blood derivatives to cell-based augmentation procedures, different lesions (ACL repair and ACLR) and studies of mixed quality [29]. The scoping review of the available literature, besides identifying a larger number of high-quality studies through a systematic search, uncovered a more complex scenario challenging the strength of previous conclusions. While a systematic review summarises the evidence about the effectiveness of a treatment by addressing specific questions with a structured process, the general purpose of conducting a scoping review is to identify and map the available evidence on a specific topic. In this study, the aim was ultimately to highlight the way the research has been conducted, by analysing and mapping the different aspects of the research revolving around PRP augmentation in ACLR. Through this critical analysis, an overall limited quality and a high heterogeneity were documented for the available studies. Furthermore, studies proved not to be suitable for an analysis aiming at quantifying specific aspects by merging different study results. Other reviews, while highlighting some of these limitations, ultimately ended

up summarising the evidence coming from merging the included publications [18, 29], which could lead, in light of the aforementioned limitations, to misleading conclusions. This scoping review builds upon those previous findings by analysing a larger number of studies, including only RCTs to ensure the highest-level quality, and critically analyses and questions the feasibility of a meaningful comparison among the published articles. In fact, a large heterogeneity was documented by mapping the literature with this scoping review with regards to all aspects concerning PRP application in ACLR surgery, with the use of different PRP products and applications modalities, as augmentation to different ACL grafts implanted with different techniques, and evaluated with a wide range of methods at different follow-ups.

PRP is an 'umbrella' terminology that includes a wide range of blood-derived products, and the analysis of the selected papers confirmed the large heterogeneity in the PRPs used for ACLR augmentation. Different preparation methods are available to obtain PRP, which can yield products with different compositions and characteristics in terms of the number of platelets, presence of leucocytes and red blood cells [38]. Although it has been already previously suggested by the literature to standardise the PRP preparation protocols and to specify its composition in detail [11], only five of the included studies documented PRP composition, which strongly limits the possibility of drawing clear indications on what can be expected with each specific PRP. Additional variables that should be considered regarding PRP properties include the volume of blood harvested and of PRP used, the revolutions per minute during the centrifugation process, the use of fresh or freeze-thawed products, the activation methods and the timing of activation with respect to the PRP application, as well as the number of applications and the time interval between ACLR and PRP augmentation [30]. Moreover, PRP application ranged from the preparation of PRF in the shape of a membrane applicable directly on the target site, to the use as a liquid product for the intra-articular injection. These different application modalities could entail different mechanisms of action and potentially different benefits in ACLR, ranging from a better maturation and integration of the graft within the bone tunnels to better joint homeostasis after surgery. Therefore, different PRP application modalities should not be considered together in a single analysis, as they could lead to misleading conclusions. Each of the aforementioned aspects characterising PRP should be addressed separately, to understand if different products may provide different results and, in the end, to optimise the use of PRP augmentation [9]. To this purpose, a new coding system has been recently proposed as a tool to help researchers and clinicians in the difficult process of delineating more precise information about PRP [30].

The use of a common reference system is needed to convey all the essential data to compare the results of different trials and guide future investigations on the application of PRP in research and clinical practice.

Another important aspect representing a bias in the comparison of the results of the different PRPs is the graft choice, which varied in these studies. In the RCTs analysed, ACLR was performed with HSs, patellar or peroneus longus grafts. Moreover, ACLR with HSs was performed with different surgical techniques with single- or double-bundle grafts. Also, surgical approaches and tunnel drilling varied across the included studies. Different fixation methods were employed, such as cortical buttons, cross pins or interference screws. While it is not the purpose of this study to elucidate the different results provided by each one of these variables, their heterogeneity and the possible influence on the PRP augmentation results should not be neglected [24, 25, 43, 54]. The substantial differences in PRP products and application modalities, together with the different graft choices, surgical techniques and fixation methods are not the only important confounders in the analysis of ACLR PRP augmentation. The focus of each study varied in terms of outcomes, as well as the methodology applied to measure them. With regard to clinical objective outcomes, different tools were used to evaluate anterior knee laxity, and they could not all provide the same reliability. For example, KT-1000 and KT-2000 arthrometers are both common methods to assess anterior knee laxity [45], but a review from Arneja and Leith [4] recommended against the use of KT-1000 as an outcome measure for ACLR. The anterior drawer test is another common method for the diagnosis of ACL rupture; however, its sensitivity ranges from 22% to 41% when it is performed in the alert patient [36], raising some concerns on its reliability along the different follow-ups. The Pivot shift test evaluation could be biased by concomitant injuries [23], and the studies were associated differently with other lesions and surgeries like partial meniscectomy or meniscal repair [5, 41]. Only two studies [31, 35] reported the Lachman test, which is considered a valuable tool to assess knee stability [15], but they stratified the results differently.

Knee laxity is a key result when performing an ACLR, although it does not necessarily reflect reconstructed ACL functional outcomes [44], neither patients' symptoms nor satisfaction, making the clinical evaluation of primary importance to evaluate the treatment success. Clinical subjective outcomes were evaluated through five different questionnaires, with follow-ups ranging from the day after surgery to 2 years. The heterogeneity of follow-ups in VAS pain scores made comparison across studies inappropriate. The VISA score, used in the study by Cervellin et al. to address the patellar harvest site healing [10], is a validated tool

for patellar tendinopathy [61]; this score is not validated for ACLR outcomes. The IKDC is considered a valid tool to address ACL surgery functional outcomes, but it was performed only by four studies and at different follow-ups, evaluating different moments ranging from the recovery phase to the final stabilised outcome.

Radiological results are also important when augmenting ACLR since PRP is supposed to improve graft integration and maturation. Even though 12 studies addressed this topic, a clear lack of consistency in the literature was found in the approach to the description and analysis of this matter [59, 60]. Different MRI techniques were employed, and the timing of MRI follow-ups was far from homogeneous: this implies that different studies investigated different graft healing phases [27], with four studies reporting follow-up results during the so-called 'proliferation phase' [35, 42, 51, 63] and eight studies reporting them during the 'ligamentisation phase' [5, 6, 19, 22, 41, 48, 52, 64]. Moreover, MRI parameters (graft signal intensity, signal-to-noise ratio and ADC) were measured at different graft locations and used to evaluate a broad variety of outcomes, such as graft maturation, ligamentisation, integration, revascularisation, microvessel permeability and vascular density. Quantification of these parameters was performed without uniform and validated scales, thus introducing a further element of heterogeneity, which prevents performing a meaningful comparison across different studies. Similarly, a clear lack of homogeneity was found among the studies that evaluated bone tunnel widening. Multiple imaging techniques were employed (MRI, CT and X-ray) [31]. The timing of imaging acquisition at follow-up varied consistently among the included studies, and some papers addressed tunnel widening only for the tibial or the femoral tunnel. Also, in this case, the relevant heterogeneity underlined by this scoping review prevents the possibility of a methodologically correct comparison across the studies.

All the above-mentioned aspects are significant limitations of the available literature. While several studies have been published and several literature analyses have been performed to summarise and quantify the benefits of PRP augmentation, this scoping analysis shed important light on the fragmentation of this field. Studies differed significantly in several aspects, ranging from the PRP product and application modality to the ACL technique in terms of graft type and fixation, to the different evaluation methods applied at different follow-up times. In this light, almost no data can be considered homogeneous, thus making any attempt to merge and summarise them at risk of introducing bias and providing misleading results, and questioning the conclusions of previous literature analyses. Overall, there is an urgent need for standardisation to improve this field and reach solid evidence on this topic. This review documented an overall poor quality of the available literature, and a limited

possibility to analyse the too heterogeneous findings, thus making it impossible to draw solid conclusions about the potential role of PRP augmentation in ACLR.

CONCLUSION

The available literature addressing PRP augmentation in ACLR is largely scattered. PRP was produced and applied following different procedures, and high variability was detected across the included studies for every aspect of ACLR surgery: surgical technique, graft choice, outcomes evaluated, follow-up time and radiological results. Currently, a meaningful comparison of the available studies is not possible as the quantification of the literature results is biased by their heterogeneity. Future studies should provide more standardisation to investigate the benefits of biological augmentation in ACL surgery and confirm the promising yet weak evidence of PRP potential as well as the most suitable application modality, before routine use in clinical practice.

AUTHOR CONTRIBUTIONS

Conceptualisation: Giuseppe Filardo and Marco Delcogliano. *Methodology:* Alessandro Sangiorgio and Alessandro Bensa. *Data Curation:* Angelo Boffa and Luca Andriolo. *Literature search and data analysis:* Alessandro Sangiorgio and Alessandro Bensa. *Writing—original draft preparation:* Alessandro Sangiorgio and Alessandro Bensa. *Writing—review and editing:* Angelo Boffa, Luca Andriolo and Giuseppe Filardo. *Supervision:* Giuseppe Filardo and Marco Delcogliano. All authors have read and agreed to the published version of the manuscript.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

ETHICS STATEMENT

Ethical approval was not required as the study was an analysis of the literature.

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